

Day 6: Advanced Power Query Transformations

[DEF] Definition to Remember

Advanced Power Query transformations allow reshaping wide datasets into normalized tall formats (Unpivot), aggregating tables (Group By), combining row sets (Append), and performing relational joins (Merge).

1. The Power of Unpivot Columns

* Problem: Financial tables often store months as separate columns (Jan, Feb, Mar...).

WIDE FORMAT (Bad for BI):

Product	Jan_Sales	Feb_Sales	Mar_Sales
Widget	100	150	200

TALL FORMAT (Ideal for Data Modeling):

Product	Month	Sales
Widget	Jan	100
Widget	Feb	150
Widget	Mar	200

2. Append vs Merge Queries

Transformation	Type	SQL Equivalent	Description
Append Queries	Vertical Stacking	`UNION ALL`	Combines rows of two or more tables with the same structure.
Merge Queries	Horizontal Join	`JOIN`	Merges columns from two tables based on a common key.

3. Group By & Custom M Columns

* Group By: Aggregates row values (e.g., Total Sales per Customer) directly during ETL.

[MUST-WRITE] Must-Master Skills & Interview Points

1. Why Unpivot is essential for dynamic time intelligence and DAX calculations.
2. Master the 6 Merge Join types in Power Query (especially Left Anti Join for finding orphan records).
3. When to aggregate in Power Query (Group By) vs calculating dynamically in DAX.

[RECALL] Quick Recall

Unpivot (Wide to Tall) + Append (Stack Rows / UNION) + Merge (Join Columns / JOIN) + Group By (ETL Aggregation)